Vertex Conflict

Two agents occupy same node at the same tick

2 x 3 grid:

0 - 1 - 2
| | |
3 - 4 - 5

Agent A: start: 0
goal: 2

Agent B: start: 2
goal: 0

A path: {0, 1, 2}
B path: {2, 1, 0}

Create two children

left: A cannot occupy 1 at tick 1
replanned path = A{0, 4, 2}; B{2, 1, 0} cost = 4

right: B cannot occupy 1 at tick 1
replanned path = A{0, 1, 2}; B{2, 4, 0} cost = 4

Both are conflict free. Need tie breaker since cost is the same. We use insertion order, choose left.

New paths: A{0, 4, 2}
B{2, 1, 0}

Insertion order is planned.
Not yet implemented.

Swap Conflict

Two agents exchange cells during same tick

2x2 grid

0 - 1
| |
2 - 3

A - start: 0; goal: 1
B - start: 1; goal: 0

A moves 0 → 1
B moves 1 → 0

conflict: swap during tick 0 → 1
total cost: 1+1 = 2

Create two children

Left constraint: A cannot move 0 → 1 starting tick 0
Right constraint: B cannot move 1 → 0 starting tick 0

Left Possible paths: A{0,0,1}; B{1,0} cost = 3
Right Possible paths: A{0,1}; B{1,1,0} cost = 3

Now have vertex conflict; waiting fixed swap conflict.
Using left branch: A{0,2,1}; B{1,0}

No more conflicts. Total cost = 2+1 = 3

Diagonal Conflict

Two diagonal movements cross during same tick

2x2 grid

0 - 1
1 X 1
2 - 3

A - start: 0 ; goal: 3
B - start: 1 ; goal: 2

A moves 0 → 3
B moves 1 → 2

Diagonals cross at the center.
Conflict: diagonal crossing at tick 0 → 1
cost = 1+1 = 2

Create two children

Left constraint: A cannot move 0 → 3 starting at tick 0
Right constraint: B cannot move 1 → 2 starting at tick 0

Possible paths:

Left: A{0,0,3}; B{1,2}; cost = 3
Right: A{0,3}; B{1,1,2}; cost = 3

Choose left: A waits while B crosses, then A crosses at next tick.

Result: no conflicts
cost: 2+1 = 3